REVIEW ARTICLE

Type-3c Diabetes Mellitus, Diabetes of Exocrine Pancreas – An Update

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ARTICLE HISTORY

Received: June 07, 2018
Revised: November 02, 2018
Accepted: January 08, 2018

DOI:
10.2174/1573399815666190115145702

Abstract: ***Background***: The incidence of diabetes is increasing steeply; the number of diabetics has doubled over the past three decades. *Surprisingly*, the knowledge of type 3c diabetes mellitus (T3cDM) is still unclear to the researchers, scientist and medical practitioners, leading towards erroneous diagnosis, which is sometimes misdiagnosed as type 1 diabetes mellitus (T1DM), or more frequently type 2 diabetes mellitus (T2DM). This review is aimed to outline recent information on the etiology, pathophysiology, diagnostic procedures, and therapeutic management of T3cDM patients.

Methods: The literature related to T3cDM was thoroughly searched from the public domains and reviewed extensively to construct this article. Further, existing literature related to the other forms of diabetes is reviewed for projecting the differences among the different forms of diabetes. Detailed and updated information related to epidemiological evidence, risk factors, symptoms, diagnosis, pathogenesis and management is structured in this review.

Results: T3cDM is often misdiagnosed as T2DM due to the insufficient knowledge differentiating between T2DM and T3cDM. The pathogenesis of T3cDM is explained which is often linked to the history of chronic pancreatitis, pancreatic cancer. Inflammation, and fibrosis in pancreatic tissue lead to damage both endocrine and exocrine functions, thus leading to insulin/glucagon insufficiency and pancreatic enzyme deficiency.

Conclusion: Future advancements should be accompanied by the establishment of a quick diagnostic tool through the understanding of potential biomarkers of the disease and newer treatments for better control of the diseased condition.

Keywords: Type 3c diabetes mellitus, insulin, glucagon, pancreatogenic diabetes, chronic pancreatitis, pancreatic ductal adenocarcinoma.

1. INTRODUCTION

Diabetes mellitus (DM) is an assorted cluster of diseases marked by chronic blood glucose level elevation. Affecting around 425 million people worldwide, diabetes is a global disease that is currently on the rise [1]. Diabetes was previously divided into three main types, namely, type 1 diabetes mellitus (T1DM), type 2 diabetes mellitus (T2DM) and gestational diabetes. However, the latest classification of diabetes mellitus by the American Diabetes Association and World Health Organization defines diabetes mellitus as a result of pancreatic diseases as pancreatogenic diabetes or T3cDM [2]. The proposed universal terminology now is diabetes of the exocrine pancreas [3], although terms such as type 3c diabetes and secondary pancreatic diabetes are still used [4]. The pancreatic diseases referred to pancreatic exocrine damage due to acute, relapsing and chronic pancreatitis (of any aetiology), cystic fibrosis, hemochromatosis, pancreatic cancer, pancreatectomy, and rare causes such as neonatal diabetes due to pancreatic agenesis [2].

There is little comprehensive information on the prevalence of T3cDM. Some older studies estimated a low prevalence of around 0.5%-1.15% among all the diabetes mellitus cases in North America [5, 6]. Separate studies from Southeast Asia, where tropical or fibrocalcific pancreatitis is endemic, show a greater prevalence of nearly 15% - 20% of all

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1573-3998/19 $58.00+.00